

Seyed Ali Rashidi

Computer Vision & ML Engineer

farid.rash@gmail.com | LinkedIn | GitHub | +393517683212 | Naples, Italy

Profile

Computer Vision & ML Engineer with a track record building end-to-end deep learning pipelines for scientific imaging — spanning medical CT/MRI segmentation, astronomical image classification, and spaceborne thermal-infrared systems. Currently Payload Supervisor on the IGNIS CubeSat mission at the University of Naples Federico II, leading SNR modelling, Zemax optical simulation, and imaging performance analysis for a thermal-IR Earth observation payload. Proficient in PyTorch, OpenCV, and Docker, with production-deployed models on Google Cloud Run. Brings rare cross-domain depth across computer vision, sensor physics, reinforcement learning, and containerized ML systems.

Skills

Languages & Frameworks — Python, C++, C, Bash, PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Plotly, Streamlit

Computer Vision & ML — Image classification, object detection, semantic segmentation, instance segmentation, CNNs, ResNet, U-Net, YOLOv8, transfer learning, hyperparameter tuning (Optuna), data augmentation, class imbalance handling, model evaluation (mAP, IoU, precision/recall, Dice)

Reinforcement Learning — DQN, Q-learning, replay buffer, epsilon-greedy exploration, reward shaping, stop-loss/target logic, PyTorch-based RL pipelines

Deployment & MLOps — Docker, REST inference APIs, Google Cloud Run, GCP, Git, CI/CD, model versioning, containerized deployment, Linux (Bash), Jupyter, MLflow

Imaging & Sensor Systems — Thermal infrared imaging, radiometric modelling, SNR/NETD analysis, microbolometric sensors (FLIR Boson 640), FoV estimation, Zemax OpticStudio, optical system simulation

Data Science & Statistics — Statistical learning, data mining, big data processing, advanced statistical modelling, time-series analysis, anomaly detection, SQLite

Professional Experience

03/2025 – Present	University of Naples “Federico II” - IGNIS CubeSat Mission Payload Supervisor — Computer Vision & Imaging Systems - Built physics-aware SNR and imaging performance models for a thermal-infrared Earth observation CubeSat payload using FLIR Boson 640 microbolometric sensor data, enabling quantitative feasibility assessment under realistic orbital conditions. - Simulated full optical system performance in Zemax OpticStudio, modelling FoV, spatial resolution, ground footprint, and signal levels across orbital scenarios. - Identified non-optimal launch timing through time-series analysis of SNR trends and seasonal orbital effects; recommended a ~5-month delay, directly improving projected early-mission image quality. - Conducted payload hazard analysis and supported engineering trade-offs in collaboration with multidisciplinary spacecraft teams. - Thesis: <i>Signal-to-Noise Ratio characterisation for thermal-infrared microbolometric imaging in CubeSat missions.</i>
01/2016 – 12/2023	EZ-Tech Data Recovery Engineer - Diagnosed and recovered data from failed storage systems across hundreds of cases, applying low-level signal integrity analysis and structured failure diagnosis under strict quality constraints. - Developed systematic recovery protocols for hardware failures, building engineering discipline in high-stakes, low-margin-for-error environments. - Managed end-to-end case workflows independently — from fault analysis to validated data delivery — consistently under time and quality pressure.

Projects

04/2026 – Present	Brain CT Hemorrhage Segmentation Intracranial Hemorrhage Detection – U-Net CT Segmentation - Built an end-to-end segmentation pipeline for intracranial hemorrhage detection from CT scans using U-Net in PyTorch. - Implemented 3D→2D CT slicing, NIfTI processing, Hounsfield Unit windowing (brain/blood/bone), and balanced patch sampling for class imbalance. - Custom loss combining BCEWithLogitsLoss and Dice Loss; deployed as a Dockerized REST API on Google Cloud Run with a Streamlit inference interface.
08/2024 – 11/2024	Radio Galaxy Classification Morphological Classification of Extragalactic Radio Sources – CNN - Designed an end-to-end CV pipeline classifying radio galaxies (FR0, FRI, FRII) from ~18,000 grayscale images (50×50px) using a custom CNN built in TensorFlow. - Plain CNN achieved best test accuracy at 82.77%, outperforming ResNet (77.81%) and Optuna-tuned variants (80.28%) – trained in under 3 minutes. - Deployed as a Dockerized REST inference API on Google Cloud Run – production-ready, scalable HTTP-based inference.
07/2025 – 11/2025	Thermal Infrared Heatmap & SNR Prediction - Built a Python-based physics-aware simulation framework to predict SNR and imaging performance for a thermal-infrared CubeSat payload under realistic orbital and environmental conditions. - Modelled signal levels as a function of orbital altitude, viewing geometry, and temperature contrast between target and background scene. - Generated thermal heatmaps and scenario-based performance distributions to evaluate imaging feasibility across mission conditions. - Conducted sensitivity analysis on ground footprint, observation geometry, and temperature differential to quantify their impact on image quality. - Outputs directly informed payload design trade-offs and mission-level engineering decisions for the IGNIS CubeSat mission.

Education

09/2023 – present Naples, Italy	University of Naples Federico II  Master’s Degree in Data Science Thesis: <i>Signal-to-Noise Ratio Characterisation for Thermal-Infrared Microbolometric Imaging in CubeSat Missions</i> Key courses: Machine Learning, Computer Vision, Statistical Learning, Data Mining, Big Data, Advanced Statistical Modelling
09/2009 – 01/2012	Islamic Azad University BSc in Computer Software Engineering Data Structures, Algorithms, Systems Design, Programming, Engineering Mathematics

Languages

English B2	Italian A2	Deutsch A1
----------------------	----------------------	----------------------

I hereby give consent for my personal data included in this CV to be processed for recruitment purposes in accordance with the European Regulation (EU) 2016/679 (GDPR).